Rapid #: -26963227

CROSS REF ID:	**58127442790003408**
LENDER:	**IUA (University of Texas at Arlington) :: Central Library**
BORROWER:	**CPV (City University of Hong Kong) :: Main Library**
TYPE:	Article CC:CCG
JOURNAL TITLE:	Proceedings of the Institution of Civil Engineers. Structures and buildings
USER JOURNAL TITLE:	Proceedings of the Institution of Civil Engineers - Structures and Buildings
ARTICLE TITLE:	Research on the intelligent construction of prefabricated building based on digital twin
ARTICLE AUTHOR:	Wen, Yafeng
VOLUME:	178
ISSUE:	9
MONTH:	
YEAR:	2025
PAGES:	851 - 863
ISSN:	0965-0911
OCLC #:	

Processed by RapidX: 6/16/2026 11:13:02 AM

This material may be protected by copyright law (Title 17 U.S. Code)

Cite this article
Wen Y (2025)
Research on the intelligent construction of prefabricated building based on digital twin.
Proceedings of the Institution of Civil Engineers – Structures and Buildings **178(9)**: 851–863,
https://doi.org/10.1680/jstbu.25.00086

Research Article
Paper 2500086
Received 13/05/2025; Accepted 25/06/2025

Emerald Publishing Limited: All rights reserved

Structures and Buildings

Research on the intelligent construction of prefabricated building based on digital twin

NOTICE: This Material May Be Protected
By Copyright Law (Title 17 U.S. Code)

Yafeng Wen
School of Intelligent Construction, Fuzhou University of International Studies and Trade, Fuzhou, PR China (wenyafeng@fzfu.edu.cn)

In recent years, the emergence of advanced information technology and artificial intelligence has significantly advanced the construction industry, leading to a new era characterized by intelligent construction. Prefabricated buildings, distinguished by their standard components, have emerged as a crucial factor in this evolution, demonstrating rapid growth. Intelligent construction embodies a groundbreaking methodology integrating advanced information technology with engineering construction techniques. An essential technical challenge involves efficiently integrating the 'physical' and 'digital' domains within prefabricated structures. This paper explores the integration of digital twin (DT) technology within the intelligent construction of prefabricated buildings. DTs offer real-time connectivity between digital and physical construction environments, supporting adaptive decision making and enhanced collaboration. The state-of-the-art literature is reviewed, a structured DT framework is proposed and applications in hoisting management, safety and predictive analytics are examined. The study provides strategic insights into advancing Construction 4.0 by addressing scientific challenges that hinder the transition to intelligent construction and establishing a theoretical foundation for this transformation.

Keywords: artificial intelligence/building information modelling (BIM)/buildings, structures & design/digital twin/intelligent construction/prefabricated building

CPV

1. Introduction

Before widespread industrialization, China's construction sector grappled with considerable challenges concerning information management and digital transformation, yet simultaneously opportunities were presented for improved quality development. Prefabricated construction, distinguished by its standardized design, industrialized manufacturing, modular assembly and comprehensive finishing, is a crucial mechanism and primary avenue for advancing new construction industrialization (Hao, 2022; Li *et al*., 2020). Achieving intellectualization within the prefabrication process necessitates real-time data exchange and multidisciplinary collaboration among all construction stakeholders, a feat made possible by advanced information technology (IT) systems (Zhong *et al*., 2017).

Recently, technologies including the Internet of Things (IoT), big data analytics, cloud computing and artificial intelligence (AI) have seen increasing adoption in the construction sector (Haris *et al*., 2024; Jagatheesaperumal *et al*., 2022; Zhong *et al*., 2017). However, their application has largely remained confined to solving isolated engineering tasks such as equipment tracking, energy monitoring and scheduling optimization (Ibrahim *et al*., 2024; Mabrouka and Younis, 2022; Wen, 2021). As a result, the industry still lacks a unified, integrated platform to bridge design, production, assembly and post-construction stages under an intelligent and data-centric umbrella.

1.1 Related works

Intelligent construction has emerged as a transformative approach that merges digital technologies with the entire construction life cycle, from planning and design to on-site management and post-occupancy evaluation (Meshref *et al*., 2023). It emphasizes dynamic decision making, real-time feedback and cross-disciplinary collaboration. Here, intelligent construction is not limited to automation alone, but extends to cognitive decision support, proactive risk control and autonomous optimization across the construction value chain (Sacks *et al*., 2020; Sepasgozar *et al*., 2023).

The digital twin (DT) – an interactive virtual replica of a physical asset or system – plays a pivotal role in enabling intelligent construction. By continuously synchronizing real-time data from sensors, devices and construction equipment with its digital counterpart, DT allows stakeholders to simulate, predict and control the physical environment in real time (Boje *et al*., 2020; Tao *et al*., 2022). While DT is widely adopted in aerospace and manufacturing, its integration in the architecture, engineering, construction and facility management (AEC-FM) sectors is relatively nascent (Hosamo *et al*., 2022; Madubuike *et al*., 2022).

Recent research, such as that by Liu *et al*. (2022b), has explored the use of DT in risk control for prefabricated hoisting operations. Similarly, Jiang *et al*. (2022) proposed a DT-enabled modular integrated construction system for on-site assembly. Although these studies illustrate the feasibility of DT, they also highlight challenges in interoperability, stakeholder alignment and scalability. In addition, Pan and Zhang (2021) presented a data-driven DT model incorporating IoT and building information modelling (BIM) for project management, showcasing how predictive analysis and dynamic simulation can optimize workflows.

Recent studies such as Mikołajewska *et al*. (2025) and Zribi *et al*. (2025) have underlined the importance of integrating Industry 4.0 technologies with DTs to address long-standing inefficiencies in the construction sector. The current author incorporates insights from these works to highlight both conceptual and practical advancements in DT-enabled automation, sustainability and real-time system responsiveness.

Despite these advances, the theoretical underpinnings of applying DT for intelligent prefabricated construction remain underexplored. Few studies have rigorously addressed how DT frameworks can align with real-world construction ecosystems to deliver scalable, interoperable and autonomous intelligent services.

1.2 Intelligent construction services

Integrating DT technology with intelligent construction services represents a paradigm shift in how buildings and infrastructure are designed, assembled and managed (Madubuike *et al*., 2022; Sepasgozar *et al*., 2023) DTs facilitate real-time monitoring, predictive maintenance, adaptive resource allocation and enhanced safety protocols (Liu *et al*., 2022a; Tao *et al*., 2022) Within this synergy, intelligent services emerge from the ability to sense, analyse and respond to conditions across various construction phases. Smart scheduling and logistics are enabled through real- time updates of site conditions (Jiang *et al*., 2022), while safety management is reinforced by way of risk modelling and early-warning systems derived from sensor data (Kor *et al*., 2023; Liu *et al*., 2021). Virtual prototyping and simulations empower designers and engineers to validate performance before implementation, minimising errors and reworking (Pan and Zhang, 2021). These capabilities promote proactive decision making and dynamic adaptability, aligning the construction process with the goals of Construction 4.0 (Hu *et al*., 2022; Kosse *et al*., 2022). In addition, integrating BIM, geographic information system (GIS), IoT and enterprise systems allows for comprehensive data exchange and inter-platform collaboration (Boje *et al*., 2020), forming the backbone of intelligent construction ecosystems.

The need to evolve prefabricated construction from isolated and manually coordinated processes into digitally synchronized and autonomous systems is more urgent than ever. Traditional construction methods suffer from fragmentation, inefficiencies and inconsistent quality control (Moshood *et al*., 2024). DT technology offers a robust framework to overcome these limitations by enabling real-time bidirectional interaction between the physical and virtual worlds (Sacks *et al*., 2020). However, several challenges remain unresolved, including the lack of scalable DT frameworks across project life cycles, the integration of intelligent services with heterogeneous construction systems, and the development of predictive models capable of learning from diverse operational data (Feng *et al*., 2021; Su *et al*., 2023). These knowledge gaps motivate this study to systematically explore how DTs can be applied not only as monitoring tools but as active enablers of intelligent, adaptive and automated construction environments. Subsequently, through consolidated interdisciplinary insights and proposing structured methodologies, the aim of this work is to lay the groundwork for future innovation in intelligent prefabricated construction.

1.3 Key contributions and organization

This article presents the following primary contributions:

- synthesis of DT technology within the context of intelligent prefabricated construction
- state-of-the-art intelligent construction services enabled by DTs, with an emphasis on safety management, automation and predictive analytics
- taxonomy of current research outcomes across multiple construction domains, highlighting common frameworks, benefits and limitations
- conceptual framework outlining the integration of DTs in prefabricated building life cycles.

CPV

The rest of the article is organized as follows: Section 2 elaborates on the core background concepts of using prefabricated construction and DTs. Section 3 focuses on using AI techniques for intelligent construction applications. Section 4 summarizes the key lessons learned and the open challenges for researchers. Finally, Section 5 concludes the article with the key inferences. Figure 1 shows the overall organization of the article.

2. Background of intelligent construction and DTs

The integration of intelligent construction with DT technology marks a significant evolution in modern construction methodologies. This section outlines the foundational concepts, technological enablers and the interplay between prefabricated construction and DT systems.

2.1 Prefabricated construction

Implications of prefabricated construction: standardized design, effective production methods and on-site assembly are all aspects of prefabricated construction that benefit from IT advancements, according to You *et al*. (2020). The approach differs from traditional building because it focuses on integration, encourages collaborative design through information sharing, uses modular assembly, highlights specialized craftsmanship, ensures precision, uses mechanized construction, coordinates decoration with the main structure, and keeps full quality management and engineering, procurement and construction (EPC) management. These elements collectively aim to optimize overall benefits throughout the construction life cycle (Torres *et al*., 2024).

Key characteristics of prefabricated structures: producing prefabricated components in a factory setting enhances energy efficiency and promotes environmental sustainability while ensuring superior

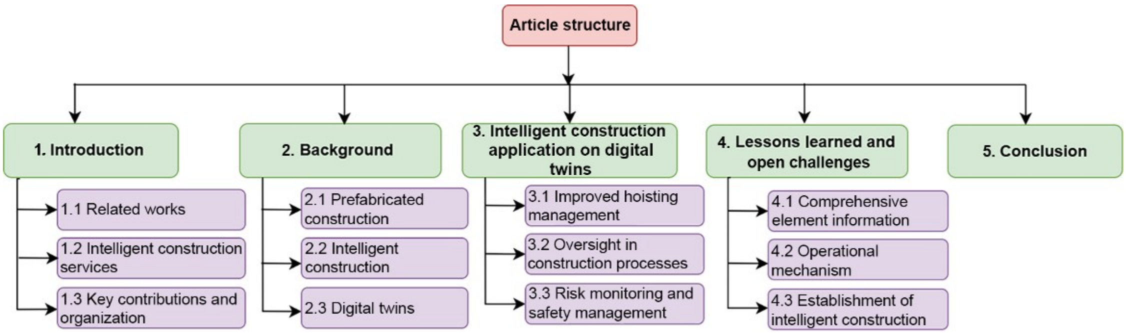
Article structure
1. Introduction
1.1 Related works
1.2 Intelligent construction services
1.3 Key contributions and organization
2. Background
2.1 Prefabricated construction
2.2 Intelligent construction
2.3 Digital twins
3. Intelligent construction application on digital twins
3.1 Improved hoisting management
3.2 Oversight in construction processes
3.3 Risk monitoring and safety management
4. Lessons learned and open challenges
4.1 Comprehensive element information
4.2 Operational mechanism
4.3 Establishment of intelligent construction
5. Conclusion

Figure 1. Overall organization of the article

quality and traceability. In addition, it allows for adaptable scheduling that aligns with the assembly's progress. The assembly of these components occurs on site by established standards, requiring only temporary support. This approach removes the necessity for peripheral scaffolding and ensures a clean and organized site, significantly reducing potential safety hazards (Ma and Liu, 2014). This methodology aligns with the principles of sustainable construction, explicitly focusing on the conservation of energy, land, water and materials alongside environmental protection.

Integrating prefabricated buildings involves electrical, structural and architectural systems, showing how design, production and assembly are all connected (Zhang *et al.*, 2023). The EPC model facilitates the management of prefabricated construction through a singular enterprise that governs the complete life cycle, encompassing design, production and assembly. The design phase encompasses the manufacturing and assembly of components and integrating decoration into the overall design process. This framework promotes the seamless integration of industrial chains while guaranteeing the construction process's structural integrity and resource optimization.

2.2 Intelligent construction

Implications of intelligent construction. Intelligent construction involves a comprehensive framework of theories, methodologies, techniques and technologies that synergize sensing, communication, digital tools, construction practices and project management. The aim of this integration is to monitor, analyse, control and optimize aspects such as safety, quality, environmental impact, progress and cost associated with construction projects (Zhang *et al.*, 2024). The notion of intelligent construction is continually evolving, particularly with the incorporation of AI and other cutting-edge technologies, resulting in increasingly innovative applications and a shifting definition (Jagatheesaperumal *et al.*, 2023a).

Key characteristics of intelligent construction. The core of intelligent construction is rooted in a thorough understanding of resource allocation in construction activities, process flow, business operations dynamics, structural health integrity, project timeline advancement and construction expense management (Jagatheesaperumal *et al.*, 2023b). The process entails gathering various and heterogeneous data, automatic analysis and learning from these data using particular strategies and algorithms, ultimately resulting in intelligent control of the construction process (Jagatheesaperumal *et al.*, 2023a).

The smart construction management platform uses sensing technology and the IoT to connect buildings, construction activities, controlled equipment, people and services (Jagatheesaperumal *et al.*, 2023c). This allows information to be sent and received interactively and in real time (Zhao *et al.*, 2021).

Closed-loop mechanisms encompass the precise detection, immediate evaluation and responsive regulation of construction operations and machinery (Liu *et al.*, 2023). Intelligent sensors can detect environmental changes, and AI technologies facilitate the analysis of extensive and varied data in real time. This process yields decision-supporting information and allows for the automatic control of production processes, accompanied by real-time feedback, as illustrated in Figure 2.

Technologies including cloud computing, fuzzy processing, optimization analysis and AI can potentially enhance decision making within the construction process. These advancements facilitate error correction, optimization, iteration and the implementation of agile warning and control systems (Mohammadi and Mukhtar, 2013). This enables ongoing enhancement and optimization of all construction processes within a closed-loop framework.

The intelligent construction system must exhibit capabilities for self-perception, adaptation, learning, decision making and execution, thereby effectively addressing complex construction challenges (Du *et al.*, 2023). The system must understand, find and change important management and resources by carefully analysing and evaluating them. It should then work as an intelligent entity that helps people and construction work together and improves the building process (Zhang *et al.*, 2024).

Integrating data flow, ubiquitous connectivity, closed-loop control and optimization empowers all construction stakeholders to

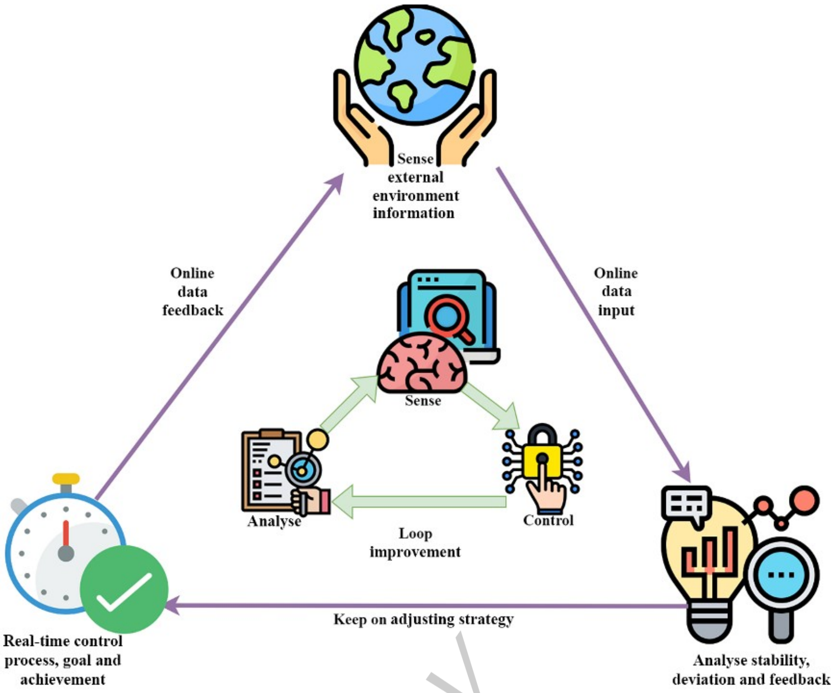
Sense external environment information
Online data feedback
Online data input
Sense
Analyse
Control
Loop improvement
Keep on adjusting strategy
Real-time control process, goal and achievement
Analyse stability, deviation and feedback

Figure 2. Loop improvement strategies for mimicking the external environments

CPV

transcend temporal and spatial limitations within the intelligent construction platform (Jiang *et al*., 2024). Such integration mitigates procedural and resource constraints, facilitating collaborative operations and enhancing transparency in management across the construction process. It exemplifies thorough and meticulous oversight, paving the way for a new era characterized by cooperative and shared production dynamics (Liu *et al*., 2022a).

2.3 Digital twins

Implications of DTs. DT technology, often called digital imaging or mapping, encompasses the virtual representation of physical products within the digital domain (Liu *et al*., 2023). It mimics digital space by combining phenomena from different fields, sizes and types of matter, accurately capturing the whole life cycle of real things (Liu *et al*., 2023).

Key characteristics of DTs. DTs create representations of the real world or models within a virtual environment, facilitating accurate and precise reconstructions and allowing for dynamic interactions between reality and these models.

DTs leverage extensive datasets, employing comprehensive analysis to classify, condense and actively oversee information (Kumar and Jagatheesaperumal, 2025). They derive insights throughout the life cycle and evaluate the worth of the data.

DTs emphasize service provision by comprehensively monitoring ‘human, machine, material, method and environment’. Their aim is to integrate energy, supply chain and resource management to enhance safety, progress, quality and cost efficiency (Jagatheesaperumal *et al*., 2023c).

DTs highlight the importance of data–model interoperability and the ability to perform transformations across multiple ports, protocols, platforms and interfaces. Such capability enhances information exchange across various industries and services, prioritizing real time and widespread connectivity.

DTs are conceptualized as representations of physical entities, highlighting the relationship between physical mapping and the digital realm. However, relying too heavily on virtual digital environments for simulations and problem solving could make it more challenging to use DT technology in real life, decreasing the likelihood that more people will accept and study it.

3. Intelligent construction application on DTs

In this section, the practical implementation of DT technology within intelligent construction is explored, highlighting its transformative impact on key operational domains. Applications in hoisting management, construction oversight, and proactive risk and safety monitoring are described.

DT technology, originally developed within the manufacturing sector’s product life-cycle management framework and first applied in aerospace, is a critical theoretical basis for the industry’s intelligent transition towards Industry 5.0 (Jagatheesaperumal and Rahouti, 2022). The emerging widespread adoption of DT applications, coupled with their demonstrated ability to integrate ‘physics’ and ‘information’, has captured the attention of various sectors eager to leverage this potential. Adapting the DT theory to improve the integration of physical and digital systems in smart buildings is important.

DTs provide a digital representation, facilitate real-time management insights and encompass a comprehensive intelligent sensing network. These systems allow real-time data integration and interactive feedback between virtual and real environments. They improve the building process by giving engineers better visual intelligence, diagnostic tools, predictive analytics and tools to help them make decisions. DTs are an important tool for combining information and physical objects, and they will help us reach our future goals for full life-cycle dynamic monitoring, visual representation and cohesive data processing in intelligent construction. The application value of DTs within this context is demonstrated in Figure 3.

Prefabricated construction, which includes collaborative design, production of parts in a factory and assembly on site with the help of reliable connection technologies, is a big step forward for the industry (Wu *et al*., 2023). From a technological application perspective, the integration of intelligent construction technologies in prefabricated buildings is frequently noted in distinct instances, such as BIM, the IoT, three-dimensional (3D) printing, AI, cloud computing and big data analytics. IT integration in prefabricated buildings is characterized by a narrow scope, fragmented approaches, and a notable deficiency in the sophistication and coherence of information exchange among stakeholders.

In the construction industry, intellectualization includes using new technologies for data mining to create a value chain that provides for data collection, transmission, computation, analysis, application and feedback (Zhao *et al*., 2021). Based on life-cycle management and lean construction principles, intelligent construction uses advanced information and construction technologies to progress from simple digitization and automation to a fully integrated and intelligent system. Such evolution promotes a construction paradigm of high quality, efficiency, low carbon dioxide emissions and safety. The different phases of intelligent construction and their associated professional fields are characterized by uniqueness and interconnectedness, requiring a strong framework for integrating physical information.

3.1 Improved hoisting management

The integration of intelligent construction methodologies and implementation of DT technology have led to notable progress in

copy

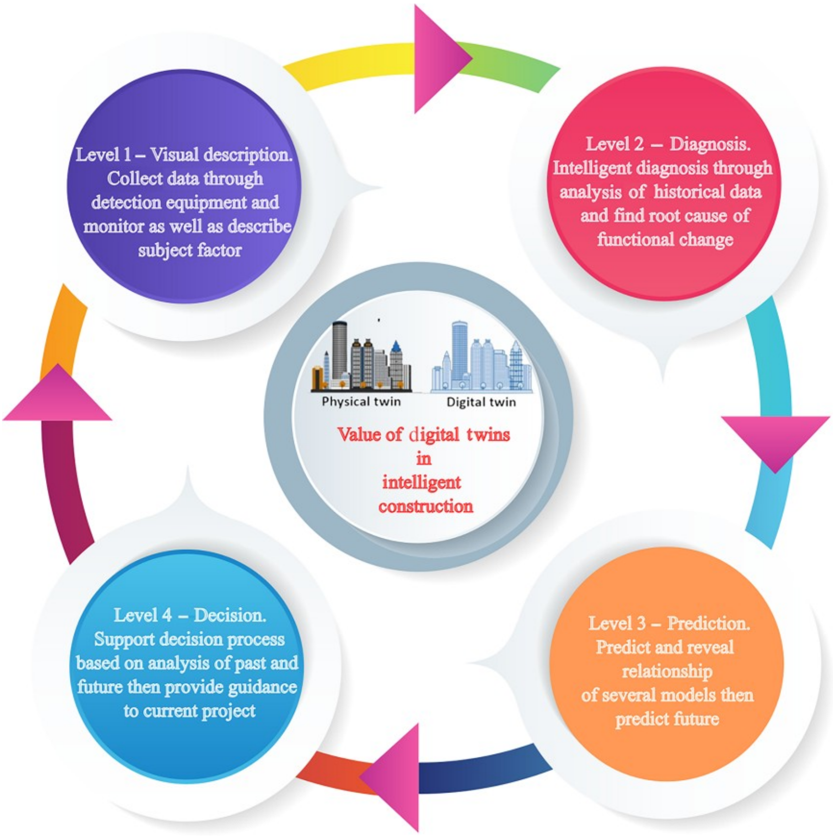
Level 1 – Visual description. Collect data through detection equipment and monitor as well as describe subject factor
Level 2 – Diagnosis. Intelligent diagnosis through analysis of historical data and find root cause of functional change
Physical twin
Digital twin
Value of digital twins in intelligent construction
Level 3 – Prediction. Predict and reveal relationship of several models then predict future
Level 4 – Decision. Support decision process based on analysis of past and future then provide guidance to current project

Figure 3. The application value of digital twins (DTs) in intelligent construction

managing hoisting operations within prefabricated building construction. The management of hoisting operations is essential in the assembly of prefabricated components. Incorporating DT frameworks has significantly improved the efficiency and safety of these processes. Implementing DT technology in hoisting management significantly increases operational efficiency and safety protocols. DT significantly improves the intelligent construction processes associated with prefabricated buildings using real-time data, predictive analytics and sophisticated decision-making tools. Several research studies have underscored the role of DT technology in enhancing hoisting processes.

Zhao *et al*. (2023) introduced a detailed framework that combines DT technology with BIM and the IoT to enhance the efficiency of hoisting management systems. The framework employs Dijkstra's algorithm to gather real-time data, strategize hoisting routes and accurately position components throughout hoisting operations. An example case study showed that this method could work in a prefabricated building project by making intelligent management of hoisting operations much better and providing a faster way to reach intelligent construction goals.

Safety remains a critical consideration in hoisting operations, as these activities can introduce considerable hazards stemming from dynamic variables encountered during the lifting and assembly of prefabricated components. Liu *et al*. (2021) explored the necessity for enhanced safety risk management by developing a DT-based safety risk coupling model. Here, the model synthesizes IoT, BIM and a complex network approach to elucidate the correlations among hoisting risk factors. The study's findings illustrate the proposed model's efficacy in improving the visualization and management of real-time safety risks associated with hoisting operations.

According to research by Liu *et al*. (2022b), developing a real-time risk prediction framework has recently improved hoisting safety. Support vector machine (SVM) algorithms added to the decision tree model make it easier for the system to predict risks related to hoisting. This facilitates the early identification of potential hazards, thereby enhancing safety protocols during the hoisting processes associated with prefabricated construction.

The findings of Liu *et al*. (2022b) emphasize the significance of DT technology in risk management during hoisting operations. The authors highlighted the capabilities of DT systems for real-time interaction, facilitating dynamic monitoring of safety risks. Adding a Bayesian network model to the DT framework creates a decision-making tool that makes risk mitigation strategies much more effective in large-scale prefabricated building projects.

Beyond prefabricated construction, various sectors, including mining, have experienced advantages from implementing DT-based hoisting management. Kalinowski *et al*. (2021) investigated the utilization of digital technologies in managing shaft infrastructure diagnostics and hoisting operations in the mining sector. The results show that DT systems can improve monitoring and diagnostics, which suggests that similar approaches could be used to improve the management of construction hoisting.

More research has been done on crane hoisting operations. For example, Jiang *et al*. (2022) created a DT-based framework to test tower cranes' stability when hoisted differently. Here, the framework provides significant insights into the safety of crane operations, highlighting essential areas for sensor deployment to mitigate unsafe behaviours during hoisting. In another similar work by Jiang *et al.* (2021), they suggested the incorporation of DT with knowledge graphs to develop a real-time early-warning system aimed at enhancing safety in hoisting operations. Here, the methodology facilitates the prompt identification of anomalies and breaches, thereby minimizing the occurrence of safety violations in construction settings.

The management of hoisting operations within industrial contexts, including mining and maritime sectors, has experienced advancements through digital technologies. Liang *et al*. (2023) introduced a DT technology system for mine hoisting, facilitating real-time monitoring and temperature forecasting of hoist motors. Such application presents significant implications for prefabricated construction, emphasizing the potential of DT technology for enhancing predictive maintenance and ensuring operational safety. In addition, Zhao *et al*. (2023) showed a DT system specially made for rockshaft hoists, showing that DT technology can be used in many fields. For example, Lipeiyong *et al*. (2024) used DT technology to improve berth hoisting planning in shipbuilding, showing how this method can improve hoisting operations in many fields.

Such recent advancements highlight the essential function of digital technologies in tackling the intricate challenges associated with contemporary construction and hoisting operations, thereby facilitating the development of more secure and efficient workflows.

3.2 Oversight in construction processes

The integration of BIM and DT technologies offers advanced simulation and management capabilities in the construction process, significantly enhancing construction oversight. Implementing DT technology facilitates monitoring and decision making in building processes, but the DT framework also presents opportunities and challenges that must be addressed for optimal use. Several studies have underscored the potential of DT to improve construction oversight by providing critical conclusions and suggestions for its application. Applying these technologies allows for early identification and resolution of construction-related issues, improving efficiency and effectiveness in building construction management (Jiang, 2021; Li *et al*., 2024). Table 1 provides a consolidated view of DT-enabled technologies enhancing hoisting management and construction oversight. It showcases various frameworks and

COPY

Table 1. Technologies and applications in improved hoisting management and oversight in construction processes

Ref.	Year	Application	Technology	Pros	Cons
Jiang *et al*. (2023)	2022	Optimized hoisting planning path	Digital twin with BIM and IoT	Optimizes hoisting efficiency	Requires complex algorithms
Liu *et al*. (2021)	2021	Real-time positioning PC	Digital twin model (DTM)	Enables accurate positioning	Needs accurate BIM data
Liu *et al*. (2022a)	2022	Risk prediction in hoisting	Digital twin-supported machine vector (DT-SVM)	Improves hoisting safety	Requires data from multiple sources
Lynch *et al*. (2023)	2021	BIM-DT integration for safety	Digital twin with BIM	Enhances decision-making	Requires full system integration
Lal *et al*. (2022)	2024	Collaboration in modular construction	Collaborative digital twin Sharing	Facilitates sharing data	Needs technology collaboration
Xiao *et al*. (2024)	2023	Automated financial management	Financial digital twin (FinDT)	Enhances financial decision-making	Requires extensive data input
Liu *et al*. (2022b)	2022	Real-time safety risk control	Bayesian network	Visualizes hierarchy risk	Difficult to build Bayesian models
Jiang *et al*. (2022)	2022	On-site modular assembly	Digital twin modular construction	Optimizes assembly workflow	Challenges in data collection
Lal *et al*. (2022)	2024	Data sharing across stakeholders	Blockchain-based digital twin sharing system	Ensures transparency	High initial setup cost
Torrecilla Garcïa *et al*. (2021)	2020	Monitoring of worker safety	IoT sensors for data collection	Enhances safety monitoring	Data privacy concerns
Pan and Zhang (2021)	2021	Prediction of workflow bottlenecks	ARIMAX time series model	Prevents process delays	Limited to task-centered projects
Liu *et al*. (2020)	2022	Improved machinery control	Augmented reality	Improves precision in operations	High cost of implementation
Hasan *et al*. (2022)	2021	Automation in design and planning	Robotic process automation (RPA)	Increases efficiency	Integration challenges

CPV

applications to improve construction operations' precision, coordination and safety.

Nguyen and Adhikari (2023) emphasized the integration of DT and BIM technologies to improve building design, performance and construction processes. They highlighted the synergies between the two technologies and how their combined use can enhance oversight by offering a real-time, data-driven approach to construction management. Nguyen and Adhikari (2023) assessed the readiness of organizations to adopt BIM-DT in sustainable construction using a structural equation modelling framework. They identified critical success factors such as cost optimization, resource management and organizational preparedness, which are essential for successfully applying DT to enhance sustainability and construction oversight.

Páldeák *et al*. (2024) highlighted the importance of accurate data validation in DT-based oversight systems. Their research emphasized using laser scanning to validate as-built data, ensuring that both geometrical and non-geometrical information of physical assets is in sync with the DT. Such a validation process is critical during the construction phase to provide stakeholders with reliable, real-time data for decision making. Sepasgozar *et al*. (2023) discussed the challenges of implementing new technologies in the construction industry. Their study presented a roadmap for future construction technologies, emphasizing the need to integrate DT with BIM and other digital technologies. Despite obstacles such as stakeholder reluctance and integration difficulties, the study underscored the transformative potential of DT, particularly for enabling remote and automated oversight capabilities.

DT has been successfully applied for real-time monitoring and predictive analysis in sectors outside construction, such as healthcare, as shown in research by Lal *et al*. (2022). These insights highlight the importance of developing DT systems that provide dynamic feedback to improve safety and efficiency during construction operations. Xiao *et al*. (2024) introduced a blockchain-based DT-sharing framework for prefabricated construction. The system encourages stakeholders to share DT resources, improving the transparency and quality of construction supervision by providing real-time data across all parties involved, especially for monitoring on-site construction and logistics.

Hosamo *et al*. (2022) emphasized the need for standardization and interoperability in DT applications within the AEC-FM industry. The lack of established standards and system integration poses a significant barrier to fully realising DT technology's potential in construction oversight, making it difficult to synthesize data from disparate systems effectively. Lynch *et al*. (2023) proposed a financial DT (FinDT) system for public sector projects, integrating financial data with project acquisition tasks. The FinDT framework enhances decision making by automating funding allocations and ensuring compliance with financial regulations, providing comprehensive oversight of project financial management and physical construction activities.

3.3 Risk monitoring and safety management

Implementing DT technology presents significant risk monitoring and safety management advancements within construction projects. It enables the utilization of real-time data alongside predictive analysis functionalities. The growing complexity of construction processes necessitates the adoption of DT frameworks, which are crucial for improving safety management practices. The progression of DT technology within construction safety management highlights its increasing influence on risk assessment and safety protocols. DT and building information modelling technologies show promise for keeping an eye on and predicting possible safety risks during the building process of prefabricated buildings (Lee and Moon, 2021). The provision of real-time data and visual management significantly enhances the analysis of safety risks and informs decision-making processes.

Liu *et al*. (2020) established a foundational framework for indoor safety management by applying DTs. It combines IoT sensors, BIM and SVMs to facilitate the automatic identification and classification of hazards within indoor settings. The framework signifies a preliminary advancement in intelligent safety management, and showcased the capability for real-time monitoring and hazard classification within a stadium context during the Beijing Winter Olympics. While concentrating on indoor safety, this study establishes a foundational framework for applying safety management practices in the construction industry. Teizer *et al*. (2022) developed a more structured approach for a DT for construction safety (DTCS), expanding upon these concepts. A workflow was delineated to improve safety by focusing on hazard prevention, risk monitoring and ongoing performance enhancement. The DTCS system prioritizes data-driven safety management throughout all project phases, offering a framework for developing future DT-based safety monitoring systems. Their vision integrates digital and physical twins, facilitating proactive hazard detection and enabling real-time feedback.

Liu *et al*. (2022a) created an intelligent safety risk prediction model based on DT technology to improve the use of DTs and risk prediction. Such a model employs a DT-SVM algorithm to forecast risks associated with hoisting in prefabricated construction. The system examines the spatial– temporal evolution of risks and delivers real-time risk alerts. Here, the advancement represents a notable progression in safety management, facilitating the anticipation of safety hazards before their manifestation and allowing for more efficient interventions. Similarly, the same authors (Liu *et al*., 2022a) introduced a more sophisticated risk control method focused on the hoisting operations of prefabricated buildings. A Bayesian network model was integrated into the decision tree framework to facilitate the visualization and analysis of the correlations among different risk factors. The model delivers immediate insights regarding safety risks, enabling construction managers to implement preventive measures effectively. Such an approach enhances decision making by establishing a hierarchy of safety risks and pinpointing the most significant risk factors. Here, the framework's effectiveness was evidenced in a large-scale prefabricated building project.

Alongside predicting safety risks, Ogunseiju *et al*. (2021) investigated the potential of digital technologies to enhance the management of ergonomic risks. The authors introduced a framework based on DT technology that uses wearable sensors to track employees' postures and an long short-term memory neural network to provide real-time feedback on ergonomic risks. The framework improves the capacity of workers to effectively manage musculoskeletal risks, leading to a decrease in injuries and an enhancement of overall well-being. Such an approach illustrates the potential for DT to transcend traditional safety management practices, incorporating tailored health and safety solutions specifically designed for construction workers.

Torrecilla García *et al*. (2021) highlighted the significance of combining BIM with DT to improve decision-making processes in occupational health and safety management. Their research emphasized the importance of collaboration in safety planning and risk awareness in construction projects, advocating for predictive risk analysis and intelligent asset optimization. Here, the integration enables a proactive safety strategy, promoting the early detection of potential hazards and enhancing communication throughout all operational tiers (Bibri *et al*., 2024). The evolution from fundamental monitoring systems to sophisticated predictive models and tailored health management illustrates how DT facilitates a more dynamic and real-time methodology in construction safety. The ongoing advancements indicate that the evolution of DT technology will significantly enhance the safety of construction environments.

4. Lessons learned and open challenges

This section reflects on critical insights gained from current applications of DTs in intelligent construction. It also identifies key technical and systemic challenges that must be addressed to advance DT-driven platforms and workflows. Table 2 presents an integrated summary of key technologies applied in risk monitoring, automation and simulation within intelligent construction contexts. It highlights the evolution of DT-driven frameworks, which enhance safety, operational efficiency and predictive analytics in modern prefabricated construction.

The core principle of intelligent construction is found in the real-time optimization and management of the construction process, reflecting a dynamic configuration production model. DT technology creates a 'digital replica' that enables the virtual reproduction of the intelligent construction process. The virtual–real fusion mechanism and interactive feedback comprise the primary cycle for sending data and information in the virtual world, as shown in Figure 4 (Torres *et al*., 2024). Throughout the life-cycle

COPY

Table 2. Technologies in risk monitoring, automation and simulation for construction

Reference	Year	Application	Technology	Pros	Cons
Liu *et al*. (2022b)	2022	Real-time risk management	Digital twin with Bayesian network	Improves safety risk visibility	Complex data modelling
Liu *et al*. (2021)	2021	Predicting safety in prefabrication	DT-SVM for safety risk prediction	Enhances risk management	Requires high computational power
Teizer *et al*. (2022)	2021	Self-management of ergonomic risks	Wearable sensors in digital twin	Reduces ergonomic injuries	High cost of wearable tech
Liu *et al*. (2020)	2022	Real-time control of construction sites	IoT and AR for real-time control	Improves on-site control	High implementation cost
Pan and Zhang (2021)	2021	Optimizing task and worker allocation	BIM-DT data mining integration	Increases efficiency	Limited to task-specific operations
Hasan *et al*. (2022)	2021	Automation in construction workflow	Robotic process automation	Reduces human error	Requires full system integration
Sacks *et al*. (2020)	2020	Predictive monitoring in construction	IoT-enabled sensors	Predicts failures early	Complex IoT set-up
Feng *et al*. (2021)	2021	Automated risk alerts	Predictive alerts through AR	Reduces human interventions	Difficult to implement in large scale
Kor *et al*. (2023)	2023	Process optimization	Dynamic workflow optimization	Enhances decision-making	Data security concerns
Kor *et al*. (2023)	2023	Integrating cognitive abilities	Deep learning and digital twin integration	Improves predictive accuracy	Requires skilled operators
Jiang *et al*. (2022)	2022	Optimizing assembly processes	Modular construction simulation	Improves assembly efficiency	High cost of technology
Sacks *et al*. (2020)	2020	Process monitoring and control	Digital twin closed-loop system	Ensures continuous feedback	High infrastructure cost
Boje *et al*. (2020)	2020	Data-driven decision making	Semantic modelling in digital twin	Improves data accuracy	Requires skilled personnel
Xia *et al*. (2022)	2022	Optimizing supply chain efficiency	Supply chain digital twin integration	Optimizes material flow	Requires advanced data models
Su *et al*. (2023)	2023	Full life-cycle simulation	Simulation of construction life cycle	Enables better forecasting	High initial setup cost

Note: AR, augmented reality

CPV

management of the building, data function as an essential conduit for transmitting construction information. To use DT technology in smart construction, four major problems need to be solved: collecting and sending all the necessary factor information; creating DT models and establishing how they work; storing and managing a lot of different types of data from different sources; and making a DT-driven platform for smart construction.

4.1 Comprehensive element information

The integration of data derived from tangible entities and digital models is crucial for thoroughly monitoring factors influencing construction production. The foundation of the DT system is built upon real-time data collection and transmission. For the virtual model to accurately reflect the operational state of physical entities, sensors must work at their best. The virtual model acts as a real-time dynamic representation of these entities. The current technical limitations affect sensors' variety, accuracy, dependability and operational context, influencing the data acquisition process. Real-time performance and security measures are at the heart of data transmission. Considering the existing technological limitations, the attainment of increased transmission speeds while maintaining network security presents significant challenges (Lin-Yao *et al*., 2019). The integration of innovative sensing methods or modules with existing systems, or their compatibility, is essential for creating a cost-effective platform that combines sensing, data acquisition and transmission, which is crucial for supporting the DT system. The data source has emerged as a critical bottleneck that impedes data analysis, utilization and the advancement of DT technology both now and in the future.

4.2 Operational mechanism

The intricate mechanism involves creating a virtual model within the digital environment, ensuring precise mapping from the physical domain to the virtual one, and achieving real-time feedback control during the construction process in the physical space.

Multi-domain modelling encompasses integrating physical systems across various domains, addressing typical and atypical conditions. Such a process spans from the initial design stage to an in-depth exploration of mechanisms, facilitating a thorough understanding and modelling of the design. Most contemporary modelling approaches are tailored to specific domains, with the integration and data fusion processes taking place subsequently to amalgamate separate models into a unified framework. Nonetheless, this fusion approach frequently proves inadequate and lacks a clear rationale, constraining integration potential. Multi-domain fusion modelling is complex because the system

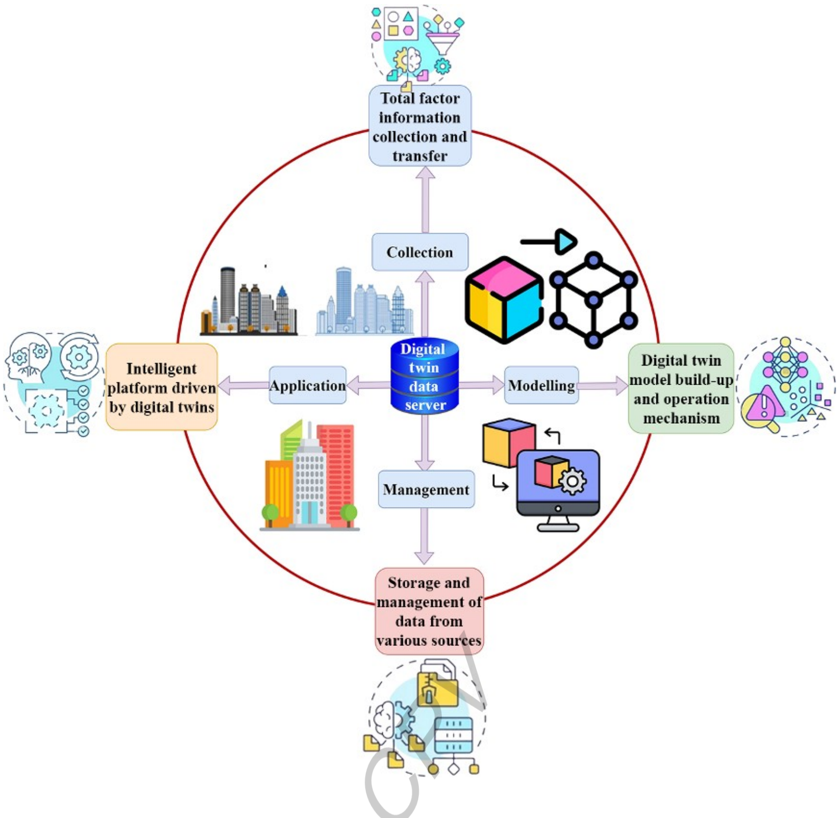
Total factor information collection and transfer
Collection
Intelligent platform driven by digital twins
Application
Digital twin data server
Modelling
Digital twin model build-up and operation mechanism
Management
Storage and management of data from various sources

Figure 4. Data and information transmission from DT server

equations have a lot of degrees of freedom that come from the different characteristics involved. At the same time, it is important for the data collected by sensors to match the actual system data precisely so that the model can be updated dynamically based on accurate sensing measurements. Three big problems are stopping the progress of DT technology: there are not enough digital design standards in the construction industry; there are not enough mathematical and simulation models to build the framework for DT technology; and, most importantly, technological processes cannot be simulated well enough digitally.

4.3 Establishment of intelligent construction

The intelligent construction system platform, which benefits from DT technology, makes making decisions and controlling the construction engineering DT's functionality easier. It could be achieved by analysing the actual needs of physical space, utilizing the virtual space algorithm base and leveraging the robust data processing capabilities of the information layer.

The intelligent construction system, as a platform that integrates various intelligent technologies, must have a technical architecture that is both interoperable and scalable. Implementing a service bus utilizes the principles of service-oriented architecture, integrating conventional middleware, extensible mark-up language and web service technologies. The construction service bus (CSB) is the essential connection point within the intelligent construction system network, enabling seamless interoperability among various services. The software subsystem in the intelligent construction system is structured as web services to obscure its internal complexity. The services offered are articulated in WSDL (web services description language) and made available through UDDI (universal description, discovery and integration). The resources involved in physical space construction, including labour and advanced construction machinery, are transformed into virtual agents and incorporated into the CSB. Such integration facilitates the distributed control capabilities of the intelligent construction system, as illustrated in Figure 5.

5. Conclusion

In this study, the implications associated with prefabricated buildings, intelligent construction and DTs are examined, focusing on the critical challenges in the intelligent construction of DT prefabricated structures. The development trends of intelligent prefabricated buildings are explored, clarifying the foundational concepts and scientific essence of intelligent construction. The key research contributions include: the identification of critical integration points between physical and digital systems; the establishment of a DT framework tailored to prefabricated construction; and the analysis of its implementation across core domains such as hoisting management, safety monitoring and predictive analytics.

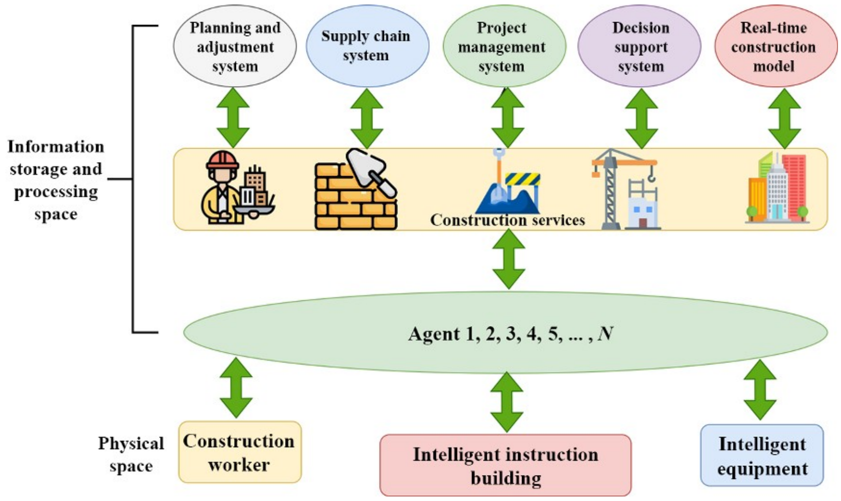
Planning and adjustment system
Supply chain system
Project management system
Decision support system
Real-time construction model
Information storage and processing space
Construction services
Agent 1, 2, 3, 4, 5, ... , N
Physical space
Construction worker
Intelligent instruction building
Intelligent equipment

Figure 5. Intelligent instruction platform integrating physical and DT space

Three important factors are discussed: gathering and sending detailed factor information; creating DT models and understanding how they work to improve the management of diverse data from multiple sources; and creating an intelligent construction platform based on DTs. Each of these factors is elaborated through a technical implementation pathway, supported by recent literature and case-based evaluations.

Practically, the findings provide actionable guidance for engineers, project managers and policymakers to enhance operational efficiency, risk mitigation and cross-stakeholder collaboration. The current study further explores the technical implementation pathways associated with each factor. Furthermore, it seeks to investigate and identify potential avenues for future research to stimulate further inquiry and practical implementations within the domain. Future work should aim to develop scalable and interoperable DT systems, expand AI-driven learning models and validate smart construction ecosystems through large-scale pilot applications.

Copy

REFERENCES

Bibri SE, Huang J, Jagatheesaperumal SK and Krogstie J (2024) The synergistic interplay of artificial intelligence and digital twin in environmentally planning sustainable smart cities: a comprehensive systematic review. *Environmental Science and Ecotechnology* **20**: 100433.

Boje C, Guerriero A, Kubicki S and Rezgui Y (2020) Towards a semantic construction digital twin: directions for future research. *Automation in Construction* **114**: 103179.

Du YL, Yi TH, Li XJ *et al.* (2023) Advances in intellectualization of transportation infrastructures. *Engineering* **24**: 239–252.

Feng H, Chen Q and de Soto BG (2021) Application of digital twin technologies in construction: an overview of opportunities and challenges. *Proceedings of the International Symposium on Automation and Robotics in Construction (ISARC 2021)*. IAARC Publications, vol. 38, pp. 979–986.

Hao L (2022) Application of interactive visualization in the prefabricated building construction process. *Acta Informatica Malaysia* **6(2)**: 52–54.

Haris M, Saad S, Ammad S and Rasheed K (2024) AI in fabrication and construction. In *AI in Material Science* (Saad S, Ammad S and Rasheed K (eds)). CRC Press, Boca Raton, FL, USA, pp. 169–192.

Hasan S, Lee K, Moon D *et al.* (2022) Augmented reality and digital twin system for interaction with construction machinery. *Journal of Asian Architecture and Building Engineering* **21(2)**: 564–574.

Hosamo HH, Imran A, Cardenas-Cartagena J *et al.* (2022) A review of the digital twin technology in the AEC-FM industry. *Advances in Civil Engineering* **2022(1)**: 2185170.

Hu W, Lim KYH and Cai Y (2022) Digital twin and Industry 4.0 enablers in building and construction: a survey. *Buildings* **12(11)**: 2004.

Ibrahim A, Nasiru J, Aminu MB *et al.* (2024) Utilising 2-d resistivity imaging tomography in investigating structural failure in some buildings in Adankolo campus of Federal University Lokoja, Nigerian north central area. *Malaysian Journal of Geosciences* **8(1)**: 59–75.

Jagatheesaperumal SK and Rahouti M (2022) Building digital twins of cyber physical systems with metaverse for Industry 5.0 and beyond. *IT Professional* **24(6)**: 34–40.

Jagatheesaperumal SK, Rahouti M, Ahmad K, Al-Fuqaha A and Guizani M (2022) The duo of artificial intelligence and big data for Industry 4.0: applications, techniques, challenges, and future research directions. *IEEE Internet of Things Journal* **9(15)**: 12861–12885.

Jagatheesaperumal SK, Bibri SE, Ganesan S and Jeyaraman P (2023a) Artificial intelligence for road quality assessment in smart cities: a machine learning approach to acoustic data analysis. *Computational Urban Science* **3(1)**: 28.

Jagatheesaperumal SK, Bibri SE, Huang J, Rajapandian J and Parthiban B (2023b) Artificial intelligence of things for smart cities: advanced solutions for enhancing transportation safety. *Computational Urban Science* **4(1)**: 10.

Jagatheesaperumal SK, Yang Z, Yang Q *et al.* (2023c) Semantic-aware digital twin for metaverse: a comprehensive review. *IEEE Wireless Communications* **30(4)**: 38–46.

Jiang Y (2021) Intelligent building construction management based on BIM digital twin. *Computational Intelligence and Neuroscience* **2021**: 4979249.

Jiang W, Ding L and Zhou C (2022) Digital twin: stability analysis for tower crane hoisting safety with a scale model. *Automation in Construction* **138**: 104257.

Jiang W, Liu Y, Chen K, Liu Y and Ding L (2023) Research on fine management of engineering projects based on BIM5D technology. In *Civil Engineering and Energy-Environment* vol. 1: 91–99.

Jiang W, Liu Y, Chen K, Liu Y and Ding L (2024) Early-warning of unsafe hoisting operations: an integration of digital twin and knowledge graph. *Developments in the Built Environment* **19**: 100490.

Kalinowski P, Długosz O and Kamiński P (2021) Digital twin of the mining shaft and hoisting system as an opportunity to improve the management processes of shaft infrastructure diagnostics and monitoring. In *Data Science, Data Visualization, and Digital Twins* (Shirowzhan S (ed.)). IntechOpen, pp. 1–10.

Kor M, Yitmen I and Alizadehsalehi S (2023) An investigation for integration of deep learning and digital twins towards Construction 4.0. *Smart and Sustainable Built Environment* **12(3)**: 461–487.

Kosse S, Vogt O, Wolf M, König M and Gerhard D (2022) Digital twin framework for enabling serial construction. *Frontiers in Built Environment* **8**: 864722.

Kumar A and Jagatheesaperumal SK (2025) Unveiling the power of digital twins in cognitive farming. In *Digital Technologies and Tools for Smart Agriculture* (Ramasamy M, Mohan K, Raj P and Sheng K (eds)). CRC Press, Boca Raton, FL, USA, pp. 170–185.

Lal A, Dang J, Nabzdyk C, Gajic O and Herasevich V (2022) Regulatory oversight and ethical concerns surrounding software as medical device (SAMD) and digital twin technology in healthcare. *Annals of Translational Medicine* **10(18)**: 950.

Lee J and Moon IY (2021) Research on digital twin automation techniques in the construction industry through 2D design drawing data extraction and 3D spatial data construction. *Proceedings of the Korean Institute of Information and Communication Sciences Conference*. The Korea Institute of Information and Communication Engineering, pp. 609–612.

Li L, Li Z, Li X, Zhang S and Luo X (2020) A new framework of industrialized construction in China: towards on-site industrialization. *Journal of Cleaner Production* **244**: 118469.

Li P, Zhao Q, Hefei Y *et al.* (2024) Research on berth hoisting planning based on digital twin. *Journal of Ship Production and Design* **40(1)**: 10–22.

Liang X, Wu J and Ruan K (2023) Simulation modeling and temperature over-advance perception of mine hoist system based on digital twin technology. *Machines* **11(10)**: 966.

Lin-Yao Y, Si-Yuan C, Xiao W, Jun Z and Cheng-Hong W (2019) Digital twins and parallel systems: state of the art, comparisons and prospect. *Acta Automatica Sinica* **45(11)**: 2001–2031.

Liu Z, Zhang A and Wang W (2020) A framework for an indoor safety management system based on digital twin. *Sensors (Basel, Switzerland)* **20(20)**: 5771.

Liu Z, Meng X, Xing Z and Jiang A (2021) Digital twin-based safety risk coupling of prefabricated building hoisting. *Sensors (Basel, Switzerland)* **21(11)**: 3583.

Liu Z, Li A, Sun Z, Shi G and Meng X (2022a) Digital twin-based risk control during prefabricated building hoisting operations. *Sensors (Basel, Switzerland)* **22(7)**: 2522.

Liu Z-S, Meng X-T, Xing Z-Z *et al.* (2022b) Digital twin-based intelligent safety risks prediction of prefabricated construction hoisting. *Sustainability* **14(9)**: 5179.

Liu Z, Zhang X, Sun Y and Zhou Y (2023) Advanced controls on energy reliability, flexibility and occupant-centric control for smart and energy-efficient buildings. *Energy and Buildings* **297**: 113436.

Lynch KM, Issa RR and Anumba CJ (2023) Financial digital twin for public sector capital projects. *Journal of Computing in Civil Engineering* **37(3)**: 04023003.

Ma Z and Liu Z (2014) BIM-based intelligent acquisition of construction information for cost estimation of building projects. *Procedia Engineering* **85**: 358–367.

Mabrouka S and Younis E (2022) Assessing construction automation and robotics in the sustainability sense. *Engineering Heritage Journal.* **6(2)**: 73–77.

Madubuike OC, Anumba CJ and Khallaf R (2022) A review of digital twin applications in construction. *Journal of Information Technology in Construction* **27**: 145–172.

Meshref AN, Elkasaby EAFA and Farid AAKM (2023) Reducing construction waste in the construction life cycle of industrial projects during design phase by using system dynamics. *Journal of Building Engineering* **69**: 106302.

Mikołajewska E, Mikołajewski D, Mikołajczyk T and Paczkowski T (2025) Generative AI in AI-based digital twins for fault diagnosis for predictive maintenance in industry 4.0/5.0. *Applied Sciences* **15(6)**: 3166.

Mohammadi M and Mukhtar M (2013) A review of SOA modeling approaches for enterprise information systems. *Procedia Technology* **11**: 794–800.

Moshood TD, Rotimi JO, Shahzad W and Bamgbade JA (2024) Infrastructure digital twin technology: a new paradigm for future construction industry. *Technology in Society* **77**: 102519.

Nguyen TD and Adhikari S (2023) The role of BIM in integrating digital twin in building construction: a literature review. *Sustainability* **15(13)**: 10462.

Ogunseiju OR, Olayiwola J, Akanmu AA and Nnaji C (2021) Digital twin-driven framework for improving self-management of ergonomic risks. *Smart and Sustainable Built Environment* **10(3)**: 403–419.

Pan Y and Zhang L (2021) A BIM-data mining integrated digital twin framework for advanced project management. *Automation in Construction* **124**: 103564.

Sacks R, Brilakis I, Pikas E, Xie HS and Girolami M (2020) Construction with digital twin information systems. *Data-Centric Engineering* **1**: e14.

Sepasgozar SM, Khan AA, Smith K *et al.* (2023) BIM and digital twin for developing convergence technologies as future of digital construction. *Buildings* **13(2)**: 441.

Su S, Zhong RY, Jiang Y *et al.* (2023) Digital twin and its potential applications in construction industry: state-of-art review and a conceptual framework. *Advanced Engineering Informatics* **57**: 102030.

Tao F, Xiao B, Qi Q, Cheng J and Ji P (2022) Digital twin modeling. *Journal of Manufacturing Systems* **64**: 372–389.

Teizer J, Johansen WK and Schultz C (2022) The concept of digital twin for construction safety. Construction Research Congress 2022, pp. 1156–1165.

Torrecilla García JA, Pardo-Ferreira M, Rubio-Romero JC *et al.* (2021) Overall introduction to the framework of BIM-based digital twinning in decision-making in safety management in building construction industry, 10.37610/dyo.v0i74.600

Torres J, San-Mateos R, Lasarte N *et al.* (2024) Building digital twins to overcome digitalization barriers for automating construction site management. *Buildings* **14(7)**: 2238.

Wang L, Chen L, Yang Z *et al.* (2022a) A prospect-theory-based operation loop decision-making method for kill web. *Mathematics* **10(19)**: 3486.

Wen Y (2021) Research on the intelligent construction of prefabricated building and personnel training based on BIM5D. *Journal of Intelligent & Fuzzy Systems* **40(4)**: 8033–8041.

Wu B, Zhao Y, Ou X, Bai Y and Zhang Z (2023) Research on fine management of engineering projects based on BIM5D technology. In*Civil Engineering and Energy-Environment* (and Gao QDuan Z (eds)). CRC Press, Boca Raton, FL, USA, vol1, pp. 91–99.

Xia L, Lu J, Zhang H, Xu M and Li Z (2022) Construction and application of smart factory digital twin system based on DTME. *The International Journal of Advanced Manufacturing Technology* **120(5–6)**: 4159–4178.

CPV

Xiao J, Ma S, Wang S and Huang GQ (2024) Fine-grained digital twin sharing framework for smart construction through an incentive mechanism. *International Journal of Production Economics* **276**: 109382.

You Z, Wu C, Zheng L and Feng L (2020) An informatization scheme for construction and demolition waste supervision and management in China. *Sustainability* **12(4)**: 1672.

Zhang R, Tang Y, Zhang Y and Wang Z (2023) Collaborative relationship discovery in green building technology innovation: evidence from patents in China's construction industry. *Journal of Cleaner Production* **391**: 136041.

Zhang L, Li Y, Pan Y and Ding L (2024) Advanced informatic technologies for intelligent construction: a review. *Engineering Applications of Artificial Intelligence* **137**: 109104.

Zhao H, Wang Y and Liu X (2021) The evaluation of smart city construction readiness in China using critic-G1 method and the Bonferroni operator. *IEEE Access* **9**: 70024–70038.

Zhao Y, Cao C and Liu Z (2022) A framework for prefabricated component hoisting management systems based on digital twin technology. *Buildings* **12(3)**: 276.

Zhong RY, Peng Y, Xue F *et al.* (2017) Prefabricated construction enabled by the internet-of-things. *Automation in Construction* **76**: 59–70.

Zribi H, Ben Abid T, Elloumi A *et al.* (2025) Industry 4.0: digital twins characteristics, applications, and challenges in-built environments. *Production & Manufacturing Research* **13(1)**: 2456277.

CPV

How can you contribute?

To discuss this paper, please email up to 500 words to the editor at support@emerald.com. Your contribution will be forwarded to the author(s) for a reply and, if considered appropriate by the editorial board, it will be published as discussion in a future issue of the journal.

Proceedings journals rely entirely on contributions from the civil engineering profession (and allied disciplines). Information about how to submit your paper online is available at www.emeraldgrouppublishing.com/journal/jstbu, where you will also find detailed author guidelines.